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Stacked Generalization Architecture for Predicting Publisher Behaviour from Highly Imbalanced User-Click Data Set for Click Fraud Detection

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Abstract

In online advertising, a change in the publisher's actual status label with every generated click shows the suspicious behaviour of the publisher. Furthermore, only a small proportion of the clicks generated by the publishers are invalid, resulting in class skewness in the dataset and a challenging issue for the conventional classification methods as they get biased towards the outnumbered class. This suspicious behaviour of publishers with an uneven class distribution ratio adversely affects the classifier's performance and increases model complexities. Thus, developing machine-learning methods capable of producing efficacious predictive models towards detecting fraudulent publishers is pivotal. This paper's novel stacked generalization framework comprises two stacked generalization architectures, one for resampling and the second for classification. The framework employs a stacked generalization approach using generalizers to improve the learning model's performance in two steps: first, reducing the error rate of algorithms towards reducing the bias in a learning set. Second, the results obtained through level-0 generalizers are fed as input to the level-1 generalizer with stacked integrated output towards combining the predictions for improving the predictive performance. Broad experimentations are conducted on FDMA 2012 user click dataset using ten-fold cross-validation. The performance of the proposed architecture is generalized by performing experiments on eight other highly imbalanced benchmark datasets, and performance is measured using average precision, recall, and F1-score. Results empirically prove the superiority of the proposed architecture in the publisher's behaviour prediction and classification as legitimate or illegitimate.

Keywords Click fraud · Class imbalance · Data sampling algorithms · Base models · Meta-model · Stacked generalization

Extended author information available on the last page of the article

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1 Introduction

The classification model's improvement on imbalanced datasets with unequal class distribution is the foremost challenge in data mining [1]. The skewed dataset severely affects several real-world classification applications such as fault diagnosis [2], financial fraud prediction [3], click fraud detection (CFD) in online marketing [4–8], etc. Online marketing or advertising is several business organisations' leading revenue source. In online marketing, the Ad-Network acts as a mediator between the Advertiser (one who creates an advertisement for ad promotion) and the publisher (advertises on websites) and holds the user's click details. Publishers are paid a proportionate amount for every user-generated click by the Ad-Network. Ad-network, in turn, earns revenue from the Advertiser for promoting ads on publisher's websites [4–8]. Publishers earn revenue using one of the widely used models of online advertising referred to as Pay-per-Click (PPC). The block diagram of the PPC model is shown in Fig. 1.

1.1 Challenging Issues of CFD

The PPC model of online advertising incentivizes publishers to generate illegitimate or false clicks on their websites, resulting in a key issue termed click fraud [9, 10]. Following are the challenges that hinder the identification of the publisher's conduct and aggravate the issue in the publisher's classification as legitimate or illegitimate:

- Million clicks generation per day
- Frequent clicks duplication per day
- Frequent changes in IP addresses
- Publisher's disguise activities
- The publisher's appearance with different labels per generated clicks
- Less amount of invalid clicks makes the class ratio uneven

Another major hitch with click fraud detection is the change in the underlying distribution of data that poses difficulty in examining the actual status label of publishers.

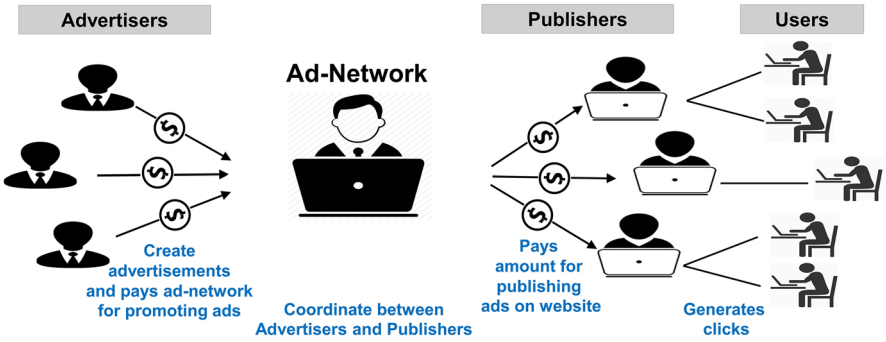

Fig. 1 PPC model of online advertising

The challenges mentioned above make the machine learning models fail to represent data characteristics [11]. In general, detecting click fraud based on analyzing publishers' behaviour depends on learning models, requiring a vast amount of features on huge datasets. Therefore, utilizing training models for look-up tends to be costly [12]. Moreover, the user's generation of millions of clicks makes the dataset massive, and a small proportion of invalid clicks out of entire generated clicks makes the dataset imbalanced. This drastic uneven class distribution in the user-click data poses major challenges to conventional learning methods, where illegitimate observations are rare compared to legitimate samples [2, 13].

1.2 Key Findings of Current Work

The current work has investigated two stacked generalizations on the imbalanced user-click dataset and other skewed datasets with resampled data: first, stacked generalization architecture for resampling (SGR), and second, stacked generalization architecture for classification (SGC). Both architectures comprise several generalizers at level-0 that fit over the training data, and whose compiled predictions are fed as an input to the meta-model at level-1 to obtain the final prediction. The first architecture fixes the issues during oversampling of the minority instances and makes the data unambiguous and balanced. The second architecture aims to seek a solution that reduces variance and, hence, averts overfitting and reduces the bias of generalizers. The stacked generalization's success arises from its capability to exploit the diversity in the level-0 classifier's predictions and thus predicts with high accuracy at level-1 (meta-level) [14]. The designed stacking architectures are appropriate as a) generalizers have significant skills on the data in distinct ways, b) the errors generated during the resampling have less correlation. The key novelties of the work are as follows:

- A novel stacked generalization framework is proposed comprising two stacked generalized architectures, designed and evaluated towards addressing the severe class skewness and other challenges that make the machine learning models fail to represent data characteristics.
- The designed architectures minimize generalizers' generalized error rate and deduce the generalizers' bias concerning providing a learning set.
- Extensive experiments are conducted on the user-click FDMA 2012 dataset with proposed architectures for publisher's identification and classification as legitimate or illegitimate.
- The performance of designed architectures is gauged by conducting experiments on eight other benchmark skewed data sets for better generalization.
- The superiority of the proposed architecture is assessed by comparing the performance with previous literature.

1.3 Previous Literature on CFD

In classification problems, a severe imbalance in the class distribution drastically affects the classifier's performance which introduces a prediction bias towards the outnumbered class. In the case where the positive class acts as a group of interest,

the false negatives prove much costlier than the false positives, which results in adverse consequences in the prediction bias of the majority class. In CFD, a greater challenge is the heterogeneity and complex structure of the user-click data sets, where the presence of legitimate publishers outnumbers illegitimate ones which make the dataset imbalance and poses complexity in publisher conduct's identification. This section thus discusses distinct works reported in the literature for the identification of fraudulent conduct of publishers for CFD in online advertising. One such work has analyzed the publisher's behaviour by conducting an empirical evaluation of major oversampling, under-sampling, and hybrid data-level resampling algorithms [15] for fair balancing of the dataset by utilizing the potentials of nine popular machine learning models. Furthermore, the utilization of oversampling technique SMOTE [7] has countered the issue of class skewness by 20% on the FDMA2012 training dataset more than other methods. Furthermore, the impact of class skewness in user click data has been analyzed by conducting experiments with actual data and data transformed using the SMOTE [16] utilizing multiple learning models. Promising results were achieved with experiments conducted with SMOTE.

Besides conventional sampling algorithms, ensemble methods such as Random Forest [17] were also used for balancing the skewed class proportions towards balancing the user-click data. Another ensemble-based model, Gradient Tree Boosting [12], handles the enormous size of user-click data in predicting the clicks and analyzing the publisher's conduct. GTB has resolved the high cardinality of categorical variables and class skewness by optimizing the loss function by averaging the outputs generated through the weak learners. Using gradient descent error minimization, GTB has outperformed other state-of-the-art models on several severely skewed datasets. In another work, by designing a quad division method [18], the size of the training dataset was reduced using the potentials of k-NN classifier by performing controlled under-sampling, where the designed procedure chose only relevant samples to make the dataset balanced. By extracting only promising samples as nearest neighbours, the designed method has reduced the time needed for generalization and has increased the classification performance. The conventional data-level over-sampling and under-sampling algorithms either overfit the data or result in data loss, while hybrid sampling algorithms compensate for both by combining both oversampling and under-sampling. One such novel hybrid data-level sampling [19] method was designed by combining SMOTE and OSS. The designed methodology has enhanced the learning model's decision-making ability while generating synthetic samples near the decision boundary while deleting the redundant samples.

Besides balancing the class distribution, selecting key features from several hundred existing features is vital for accurately identifying and classifying fraudulent publishers. Such relevant features were investigated by designing a novel feature selection technique based on majority voting [20]. The feature distillation procedure of the designed methodology has elected the significant features employing the majority-voting-based procedure utilizing the potential of popular filter and wrapper-based variable elimination methods. In contrast, the accumulated selection of the methodology has enumerated the accumulated evaluation of the feature set to obtain a subset of optimal features for the better investigation of fraudster's conduct. Since the publisher's behaviour was dynamic, significant optimal features could not detect several distinct disguise activities

of publishers. The issue was resolved by designing eight novel conditional features [21] by concatenating the multiple attributes over fined-grained periods. The designed features better captured the changing behaviour of fraudulent publishers.

Besides designing new features, the distinction between legitimate and illegitimate publishers was investigated with the help of automated feature extraction methods. For automated feature extraction, a deep CNN-based transfer learning framework [22] was designed to transform the user-click non-image features into 2D images and then use the pre-trained CNN models as effective feature extractors. Features extracted by these feature extractors helped identify the publisher's conduct. Moreover, the extraction of hidden features from an ensemble architecture [23] was reported in the literature, comprising CNN and BiLSTM. The designed hybrid deep learning-based model has automatically extracted features from the clicks generated by publishers and classified them using the Random Forest model to distinguish fraudulent publishers among legitimates. Furthermore, a blockchain-based methodology [24] was designed to detect and prevent click fraud. The designed methodology has accurately detected and prevented the clicks generated by the bots or humans.

Limitations from the literature: We reviewed the current literature towards identifying and classifying fraudulent publishers, where the rare presence of fraudsters results in skewed class distribution. During investigating the underlying key towards fraudsters' identification in user-click data, it has also been observed that the data is not instinctively systematical. Furthermore, it has also been examined from the literature that an optimum data-resampling strategy has not been investigated yet, which might help the machine learning classifiers analyse the publisher's behaviour. On the contrary, our current work addresses the abovementioned issues by proposing two stacked generalization architectures where first, resamples the dataset and another classifies the illegitimates from the legitimate ones. The comparison of the proposed methodology with the works reported in the literature for CFD is demonstrated in Table 1.

The rest of the work is structured as follows. Section 2 briefly discusses the data acquisition details. Then, the proposed architecture is elaborated in Sect. 3. The evaluation measures are discussed in Sect. 4. Experimental results of stacked generalization with resampled data are elaborated in Sect. 5. Finally, the discussion and the concluded work is presented in Sect. 6 and 7, respectively.

2 Data Acquisition

Extensive experiments are conducted on Fraud Detection in Mobile Advertising (FDMA2012) [25] user-click data for the publisher's identification and classification. The dataset is provided in CSV format and is further divided into two datasets: (i) click dataset, which keeps the click details, and (ii) publisher dataset, which keeps the details of the publisher corresponding to every publisher. The status of the publisher is defined as OK (legitimate), Fraud (illegitimate), and Observation (neither legitimate nor illegitimate, i.e., under observation). The detailed click statistics of the dataset are demonstrated in Fig. 2. The sample of the publisher and

Table 1 Comparison of the proposed methodology with various methodologies discussed in the literature for detecting click fraud

S.no	Authors	Years	Data set used	Methodology used	Remarks
1	[7]	2013	FDMA2012	Random forest	Advantage: the designed model has demonstrated generalizability to distinct datasets Disadvantage: didn't examine the features by concatenating multiple attributes identifying fraudulent behaviour of publishers
2	[12]	2020	FDMA2012	GTB	Advantage: effective classification of fraudulent publishers was performed by GTB while enhancing the performance Disadvantage: the issue of skewed class distribution is not addressed
3	[15]	2021	FDMA2012	ADASYN with GTB	Advantage: oversampling techniques outperformed other sampling methods towards enhancing the classification results Disadvantage: existing features were used for investigating fraudsters' conduct
4	[16]	2015	FDMA2012	RFE and HDDT	Advantage: newly designed features helped in modelling the behaviour of publishers Disadvantage: existing features were used for investigating fraudster's conduct
5	[17]	2012	FDMA2012	Random forest	Advantage: the analysis revealed significant discrepancies among publishers' predicted and correct status labels Disadvantage: lack of ground truth poses difficulty in the publisher's identification
6	[18]	2022	FDMA2012	QDPSKNN	Advantage: enhanced classification performance while reducing the training set size Disadvantage: not suitable for datasets containing noisy instances, outliers and missing values
7	[19]	2022	FDMA2012	SMOTEOSS	Advantage: maximized classification performance with reduced overfitting Disadvantage: random selection of samples results in the conception of useless instances
8	[20]	2022	FDMA2012	FDAS	Advantage: enhanced relevancy with reduced redundancy Disadvantage: higher computational time
9	[21]	2023	FDMA2012	Bagging + DT	Advantage: newly designed conditional features have better captured the dynamic behaviour of publishers Disadvantage: significant features must be investigated for efficacious identification of fraud publishers'
10	[22]	2022	FDMA2012	DCNNTr	Advantage: no manual feature extraction is required Disadvantage: huge memory requirements
11	[23]	2022	Talking data	CNN-BiLSTM	Advantage: automated feature extraction has enhanced classification performance Disadvantage: feature importance needs to be identified
12	[24]	2022	User-clicks	Blockchain	Advantage: the methodology significantly averts clicks generated by machines Disadvantage: higher computational time

Table 1 (continued)

S.no	Authors	Years	Data set used	Methodology used	Remarks
13	Proposed		FDMA2012	Stacked generalization architecture	Advantage: the proposed architecture appropriately balances the data as it utilizes the skills of different potential oversampling methods in distinct ways Disadvantage: the training time taken by stacked models is higher than other state-of-art models and needs more memory

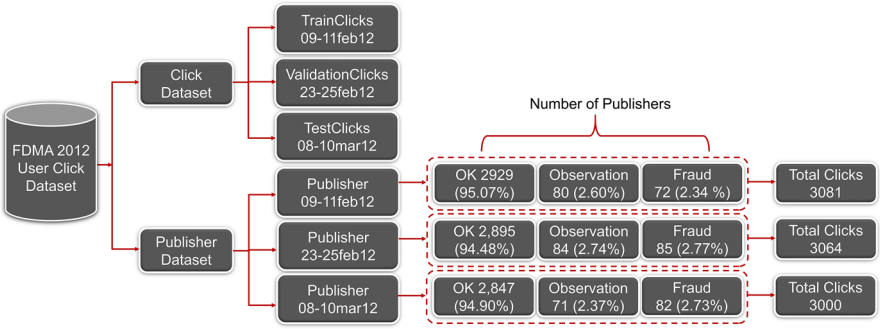

Fig. 2 Summarized click statistics of FDMA2012 user click dataset

Table 2 Instances of publisher dataset

S.no	Partner_id	Bank _account	Address	Status
1	8jj9d	?	1kp62l160if4w8cc	OK
2	8izt5	41ro6j22m3ms4w4o	?	Observation
⋮	⋮	⋮	⋮	⋮
3081	⋮	⋮	⋮	⋮

click datasets are shown in Tables 2 and 3 respectively, comprising missing values in bank_account, address, referrer and agent attributes.

The high cardinality of the dataset is resolved for better data representation. Every attribute of the dataset is analyzed and assessed for its impact on the behaviour of publishers. The extraction of significant features of high relevance is performed using distinct statistics including average and standard deviation at different periods. Overall 103 features are extracted from the FDMA2012 dataset for examining the publisher's conduct.

3 Proposed Stacked Generalization Architecture

It is argued that stacked generalization should maximize generalization accuracy rather than a single generalizer [26]. This work proposes stacked generalization; an approach aiming to obtain a generalization accuracy (contrary to learning accuracy) that is as

Table 3 Instances of click dataset

S.no	Id	Iplong	Agent	p_id	c_id	cn	Timeat	ca	Referer
1	3,036,123	700,371,228	Nokia_2626	8jljr	8ke8p	ng	00:00.0	es	?
2	3,036,121	3,673,743,995	?	8iqzc	8gspj	in	00:00.0	es	365wsqn3c-1s0ko4c
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1,048,576	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

high as possible. The main principle of stacked generalization [27, 28] is to concatenate several model predictions formatted with a learning method for improving the robustness of an individual model. Stacking includes employing models towards learning to integrate the predictions achieved from ensemble members effectively. The stacking architecture comprises 2 or more base learning models, often termed Level-0 models and a meta-model termed as a Level-1 model.

- Generalizers/base models (level-0 models): base models are those learning methods that fit over the training data, and whose predictions were compiled.
- Generalizer/meta-model (level-1 model): the meta-model is a learning method that best combines the predictions generated by the base learning models.

The process of stacked generalization is as follows:

Let the training dataset be $D = \left\{\left(x_n, y_n\right), n = 1, \ldots N\right\}$, here, x_n denotes the vector demonstrating the samples of $n$thinstance and y_n represents the target value, randomly splitting the dataset into J equal parts $D_1 - D_j$. Defining D_j and D^{-j}=$D - D_j$ as the train set and test set to the jth fold of J-fold cross-validation. Considering K base learning methods (Level-0 Generalizers), evoke the $k$thmethod over the training set D^{-j} towards inducing a learner ${M_k}^{-j}$, $k = 1, \ldots, K$. These models are termed Level-0 Models.

For every sample x_n in D_j, the testing set for the jth fold, the prediction generated by the model ${M_k}^{-j}$ on a sample x_n is denoted by z_{kn}.

$$z_{kn} = {M_k}^{-j}(x_n) \tag{1}$$

Upon completion of the process of cross-validation of every classifier, the data compiled from the outcomes of the K models are-

$$D_{cv} = \left\{\left(y_n, z_{1n}, \ldots, z_{Kn}\right), n = 1, 2, \ldots, N\right\} \tag{2}$$

where D_{cv} indicates the training dataset provided to the Level-1 model. Employ a learning method (Level-1 Generalizer), towards deriving a learning model $\widetilde{M_{eta}}$ from the data towards y as a function of $z_1, \ldots, z_K$. This is termed as Level-1 Model. Figure 3 demonstrates the procedure of cross-validation. Towards completing the training process, the final Level-0 models $M_k = 1, ..., K$ are received employing entire data in D.

Now consider the process of classification, using the models M_k, $(k = 1, \ldots, K)$ together with Meta learner $\widetilde{M_{eta}}$. With a new sample, models M_k generates a data vector as $(z_1, \ldots, z_K)$. This data vector is fed as an input to the Meta-model $\widetilde{M_{eta}}$ at Level-1, whose outcome is the final classification result for a sample.

In this framework, two stacked generalization architectures are designed, one for resampling the uneven data distribution and the second for classification predictive problems. A visual illustration of the proposed Stacked Generalization Architecture designed in this framework is presented in Fig. 4 and discussed in Algorithm 1. Following are the two proposed architectures:

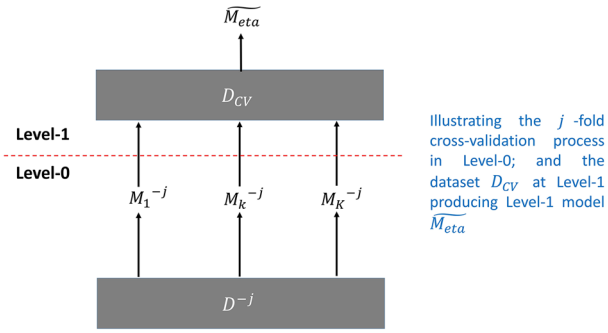

Fig. 3 Cross-validation process of the general stacked generalization procedure

3.1 Stacked Generalization Architecture-I: *Stacked Generalization for Resampling (SGR)*

Skewness in the data makes the predictive model biased and inappropriate, which degrades a learner's performance. As the minority class is more prone to error than the majority class towards detecting click fraud, it is thus essential to identify a legitimate user as legal and an illegitimate user as illegal. Therefore, a stacked generalization resampling (SGR) architecture is designed and evaluated to balance the data distribution with unambiguous instances. SGR uses base models as level-0 models and a meta-model that integrates base-model's resampling, referred to as the level-1 model.

3.1.1 Level-0 Generalizers

As base models, five major popular oversampling algorithms are used, namely, SMOTE [29], BorderlineSMOTE-1[30], BorderlineSMOTE-2 [30], SVMSMOTE

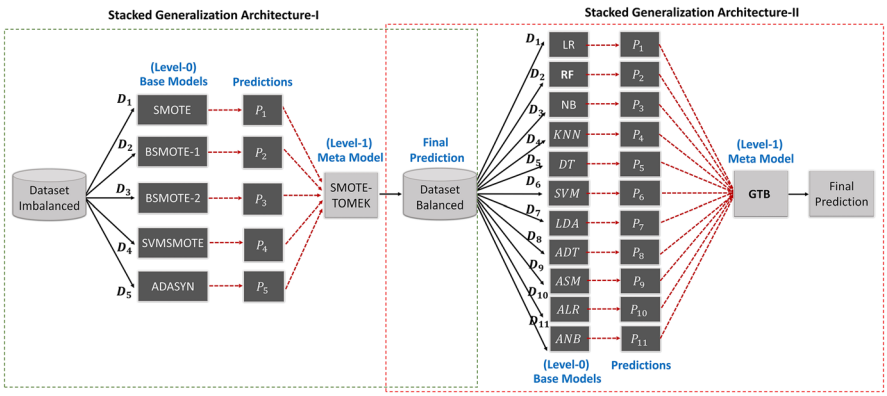

Fig. 4 Visual illustration of stacked generalization architecture

[15] and ADASYN [15]. These base models fit the training data to oversample the minority class instances, which are then compiled by the meta-model. For each of these sampling methods, we now discuss the formula used by the methods to generate new data used as input for the next level.

- SMOTE: SMOTE generates the new synthetic instance S_{new} based on the formula discussed in the following equation:

$$S_{\text{new}} = x_i + \lambda \times \left(x_{zi} - x_i\right) \tag{3}$$

where λ is the random number of range [1], x_i is the minority sample, x_{zi} is the nearest neighbor of the same class.

- The new instances $BS_{1\text{new}}$ generated by BorderlineSMOTE-1 on the basis of the formula discussed in the follow equations:where P'_i is the borderline instance of the minority class P, r_j is the random number between 0 and 1, $dif_{j,}$ is the difference among P'_i and its s nearest neighbours from P

$$BS_{1\text{new}} = P'_i + r_j \times dif_{j,}\ j = 1, 2, \ldots, s \tag{4}$$

- Borderline-SMOTE2 creates a new sample $BS_{2\text{new}}$ near to the decision boundary using the nearest-neighbors of similar class and vice-versa. The difference amid it and its closest negative neighbor is multiplied by a random number amid 0 and 0.5; thus, the newly generated samples are nearer to the minority class. The formula to generate instances by BorderlineSMOTE-2 is discussed in the following equation:where r_j = random number between 0 and 0.5, x_{zi} is the nearest neighbor of any class.

$$BS_{2\text{new}} = P'_i + r_j \times dif_{j,}\ j = 1, 2, \ldots, s \tag{5}$$

- The SVMSMOTE randomly generates the new synthetic sample along with the line segment joining x_i and x_{zi} employing interpolation and extrapolation methods, depending upon the density of the majority class samples around x_i. Interpolation generates new data samples between two instances, while extrapolation generates new data samples outside of the two instances. The new instance SM_{new} generated by SVMSMOTE is discussed in Eq. 6 as:

$$SM_{\text{new}} = x_i + \lambda \times \left(x_{zi} + x_i\right) \tag{6}$$

- ADASYN uses the same approach towards generating new samples as SMOTE does. The difference between them is that considering the density distribution, ADASYN decides the number of synthetic samples generated for instances that are difficult to learn. Thus, it helps in changing the decision boundary based on the instances that are difficult to learn. The formula used by ADASYN to generate a new instance AD_{new} is shown in the following equation

$$AD_{new} = x_i + \lambda \times \left(x_{zi} - x_i\right) \tag{7}$$

3.1.2 Level-1 Generalizer

- SMOTETOMEK [15] is employed as a meta-model that best combines the predictions generated by the base learning models and generates a final prediction. In other words, the task of SMOTETOMEK is to learn to combine best the resampling performed by the base models. After combining the instances, SMOTETOMEK then identifies and removes the majority and minority instances, making the tomek links. A Tomek link amid two samples of distinct classes *xandy* is defined for any instance z as shown in the following equation:

$$d(x, y) < d(x, z) \text{ and } d(x, y) < d(y, z) \tag{8}$$

where, $d(.)$ represents the distance amid the two samples. Tomeklink exists when two instances are the near-neighbour of each other.

3.2 Stacked Generalization Architecture-II: *Stacked Generalization for Classification (SGC)*

3.2.1 Level-0 Generalizers

Eleven standard machine learning classifiers (generalizers), namely, LR [31–33], RF[25, 34, 35], k-NN [36, 37] [18], DT(C4.5) [38, 39], NB [40], SVM [41–44], LDA [45, 46], AdaBoost + DT (ADT) [12], AdaBoost + SVM (ASM) [12], AdaBoost + LR (ALR) [12], AdaBoost + NB (ANB) [12] are used as base models. These base models are fit over the training data, and compiled results are then fed to the meta-model.

- Logistic Regression (LR): It is an approach that models the relationship between several independent variables and dependent variables. Rather than fitting a straight line, the LR employs a logistic function towards squeezing the linear equation's output amid 0 and 1as:

$$\text{Logistic}\,(\eta) = \frac{1}{1 + \exp(-\eta)} \tag{9}$$

- Random Forest (RF): RF generate predictions for unobserved instance x' by taking the average of the predictions from entire regression trees on x'.

$$\hat{f} = \frac{1}{B}\sum_{b=1}^{B} f_b(x') \tag{10}$$

Where B = B times repeated bagging, f_b = classification tree, x' = unobserved instance.

- k-Nearest Neighbor (k-NN): For a specific query sample f_t employing the majority voting [47, 48], k-NN is defined as:

$$c_t = \underset{z \in \{z_1, z_2, \ldots, z_p\}}{\arg max} \sum_{f_i \in M(f_t, k)} E(c_i, \mathrm{z}) \tag{11}$$

Where c_t = predicting class for the query sample (f_t), z= distinct class's set $z_1, z_2, \ldots, z_p$ where p= all classes, $M\left(f_t, k\right) = setofk - \text{prototypesnearestto}f_t$

- DT (ID3): With data of N attributes, DT employing 'Entropy' decides which attribute should be placed at the root or a distinct level of a tree as an internal node as:

$$E(S) = \sum_{i=1}^{c} -p_i \, log_2 \, p_i \tag{12}$$

- Naïve Bayes (NB)**:** Naive Bayes computes the posterior probability $P(c|x)$ from $P(c), P(x), P(x|c)$ as

$$P(c|x) = \frac{P(x|c)\,P(c)}{P(x)} \tag{13}$$

Where, $P(c|x)$ = posterior probability, $P(x|c)$ = likelihood, $P(c)$ = class's prior probability, $P(x)$ =predictor's prior probability.

- Support Vector Machine (SVM): Optimizing the given primal problem, SVM seeks a hyperplane to maximize the margin amid distinct classes as:

$$\min_{w,b,\xi} \frac{1}{2} w^T w + C \sum_{i=1}^{N} \xi_i \tag{14}$$

$$y_i\left(w^T \varnothing\left(x_i\right) + b\right) \geq 1 - \xi_i, \;\; \xi_i \geq 0, i \in 1 \ldots ..N$$

where w= hyperplane's normal vector, b = offset of a plane from its origin, ξ_i = slack variable employed to regularize against over-fitting, and C= soft margin constraint to penalize misclassified instances.

- Linear Discriminant Analysis (LDA): LDA provides an optimal solution by predicting points from a second class if the log of a likelihood ratio is higher

than threshold T as:Where $(\overrightarrow{\mu_0}, \sum_0)$ = the normal distribution with mean, $(\overrightarrow{\mu_1}, \sum_1)$= the normal distribution with covariance parameters.

$$(\vec{x} - \overrightarrow{\mu_0})^T \sum_0^{-1} (\vec{x} - \overrightarrow{\mu_0}) + ln\left|\sum_0\right| - (\vec{x} - \overrightarrow{\mu_1})^T \sum_1^{-1} (\vec{x} - \overrightarrow{\mu_1}) + ln\left|\sum_{01}\right| > T \tag{15}$$

- AdaBoost: AdaBoost is a boosting-based ensemble method combining multiple weak classification models f_t(DT, SVM, LR, NB) [12] into a robust learning model. Where f_t = weak model taking x as input and returning a value denotes the class of an object.

$$\mathrm{F}_T(x) = \sum_{t=1}^{T} f_t(x) \tag{16}$$

3.2.2 Level-1 Generalizer

GTB [25, 49] is used as a meta-model at Level-1, which provides a smooth interpretation of the predictions generated by Level-0 models.

- Gradient Tree Boosting (GTB): GTB aims to search for a function $\widehat{F}(x)$ for approximating the output variable from the input variable, which is formulated by minimizing a loss function $L(y, F(x))$ as.

$$\widehat{F} = \arg\min_F E_{x,y}\left[L(y, F(x))\right] \tag{17}$$

where $L(y, F(x))$=loss function

The following are the benefits of proposed stacked generalized architectures:

(i) Our proposed stacked generalized architectures learn the reliable models using the meta-model, which best combines the base model's output. At the same time, conventional methods like voting do not clarify which model to trust.
(ii) The advantage of the stacked generalization resampling (SGR) approach is that it harnesses the abilities of a range of well-performing oversampling techniques for data balancing that performs superior to an individual oversampling method.
(iii) The oversampling methods used in the architecture generate synthetic new instances in one similar direction by connecting two diagonal examples over a single line, which complexes the decision surface. The proposed approach overcomes this issue by utilizing SMOTETOMEK at Level-1, which identifies and removes those majority and minority instances that makes tomeklinks. Thus, the designed architecture balances the dataset while removing noisy or ambiguous samples.
(iv) The proposed architecture appropriately balances the data as it utilizes the skills of different potential oversampling methods in distinct ways. This is

because the base models perform resampling so that the issues that occurred during the resampling have a minimum correlation.

Algorithm: Stacked Generalization Architecture

Input:

$D =$	$\{(x_n, y_n), n = 1, \dots N\}$	// Training data (imbalanced)
$K =$	Number of base learning models of SGR and SGC	
$RM_k^{-j} =$	$k = 1, \dots, K$	// Level-0 models of SGR
$\widetilde{RM_{eta}} =$	Meta Learner	// Level-1 model of SGR
$RD_{cv} =$	$\{(y_n, z_{1n}, \dots, z_{Kn}), n = 1,2, \dots, N\}$	// The predicted training dataset generated by the level-0 models and provided to the Level-1 model of SGR
$f_t =$	$\{(x_n, y_n), n = 1, \dots N\}$	// Training data (balanced) obtained from the SGR
$CM_k^{-j} =$	$k = 1, \dots, K$	// Level-0 models of SGC
$\widetilde{CM_{eta}} =$	Meta Learner	// Level-1 model of SGC
$CD_{cv} =$	$\{(y_n, z_{1n}, \dots, z_{Kn}), n = 1,2, \dots, N\}$	// The predicted training dataset generated by the level-0 models and provided to the Level-1 model of SGC

Output: s_t

Method: *// Stacked Generalization Architecture-I (SGR) //*

Step-1: Learning Level-0 models of SGR
for $k \leftarrow 1$ *to* K do
Training the Level-0 base learning models of SGR on dataset D
$h_t = RM_k^{-j}(D)$
end for
Step-2: Constructing the training dataset for Level-1 model of SGR
for $i \leftarrow 1$ *to* m do
Constructing the training dataset to provide to the Level-1 model of SGR
$RD_{cv} = \{(y_n, z_{1n}, \dots, z_{Kn}), n = 1,2, \dots, N\}$
end for
Step-3: Learn the meta-model $\widetilde{RM_{eta}}$ of SGR on the predicted dataset RD_{cv} obtained from Level-0 models
$f_t = \widetilde{RM_{eta}}\,(RD_{cv})$
// Stacked Generalization Architecture-II (SGC) //
Step-4: Learning Level-0 models of SGC
for $k \leftarrow 1$ *to* K do
Training the Level-0 base learning models of SGC on the dataset f_t
$p_t = CM_k^{-j}(f_t)$
end for
Step-5: Constructing the training dataset for Level-1 model of SGC
for $i \leftarrow 1$ *to* m do
Constructing the training dataset to provide to the Level-1 model of SGC
$CD_{cv} = \{(y_n, z_{1n}, \dots, z_{Kn}), n = 1,2, \dots, N\}$
end for
Step-6: Learn the meta-model $\widetilde{CM_{eta}}$ of SGC on the predicted dataset obtained from Level-0 models
$s_t = \widetilde{CM_{eta}}\,(CD_{cv})$
return s_t

4 Evaluation Metrics

Though several methods are reported in the literature for evaluating the learning approaches using accuracy, this is not appropriate in unequal class distribution. Therefore, towards countering the imbalanced issue of the dataset, accuracy and

Table 4 Key measures used for the evaluation of skewed user-click dataset

Measures	Description	Equations	Eq.no
Average precision (AP)	The ratio of accurately classified relevant samples among overall actual predicted instances [51, 52]	$AP = \frac{\sum_{i=1}^{l} \frac{t_{p_i}}{t_{p_i}+f_{p_i}}}{l}$	(1)
Sensitivity/recall (RE)	A percentage of accurately classified relevant samples [53]	$RE = \frac{t_{p_i}}{t_{p_i}+f_{n_i}}$	(2)
F-measure (F1)	A harmonic mean of AP and RE [54]	$F1 = \frac{2t_{p_i}}{2t_{p_i}+f_{p_i}+f_{n_i}}$	(3)

error rate could not be considered as key measures for evaluating the classification performance due to the majority class bias while avoiding the minority class. Therefore, Average-Precision (AP), Sensitivity/Recall (RE), and F1-measure (F1) are used as evaluation measures [50] and are discussed in Table 4.

5 Experimental Results

Experiments are implemented using Python version-4 using numpy [55], scipy [56], sklearn [57], imblearn [58], and pandas [59], matplotlib [60], csv [61] libraries. In 1st architecture, five popular oversampling methods and an individual hybrid sampling technique are used as level-0 and level-1 generalizers. On the contrary, in the 2nd architecture, 12 standard machine learning classifiers are employed as base models and meta-learner. The section first represents the benchmark datasets used in the experimentation, the parameters used to tune sampling methods, and experimental results.

The architecture is designed to address the challenges confronted in the FDMA2012, a user-click dataset to identify and classify the publishers as honest or dishonest in online advertising. The generalized architecture behaviour is evaluated by performing experiments on eight other datasets obtained from KEEL and other repositories.

5.1 Dataset Details

Table 5 briefly explains the details of benchmark datasets used in the experiment. Datasets are collected from different sources and organized into binary and multiclass, with distinct characteristics. For example, the dataset includes distinct features, a number of instances, and containing no missing values. All the datasets are pre-processed except the FDMA, which needs to be pre-processed before performing the experiments. During pre-processing, features are extracted manually, and a feature vector of size 2523×103 is designed for experimentation.

Table 5 Dataset details

Datasets	Source	Category	Features	Instances	Classes	Missing values
FDMA 2012 [D1]	FDMA 2012 competition [62]	Multi class	103	2523	03	No
Satimage [D2]	KEEL [63, 64]	Multi class	36	6435	06	No
Forest Cover [D3]	KEEL [63, 65]	Multi class	54	581,012	07	No
Balance [D4]	KEEL [63, 64]	Multi class	4	625	03	No
Oil Spill [D5]	MKLab [66]	Binary	49	937	02	No
Pima [D6]	KEEL [63, 65]	Binary	8	768	02	No
Phoneme [D7]	KEEL [63, 67]	Binary	5	5404	02	No
Yeast1 [D8]	KEEL [63, 67]	Binary	8	1484	02	No
Wisconsin [D9]	KEEL [63, 67]	Binary	9	683	02	No

Table 6 Parameter setting of data sampling algorithms

Methods	Name	Sampling strategy	Randomization	Other parameters
Over-sampling	SMOTE	Not majority	Random_state = 42	k_neighbors = 5
	BSMOTE1	Not majority	Random_state = 42	Kind = borderline-1
	BSMOTE1	Not minority	Random_state = 42	Kind = borderline-2
	SVMSMOTE	Not minority	Random_state = 42	k_neighbors = 5
	ADASYN	Not minority	Random_state = 42	n_neighbors = 5
Hybrid-sampling	SMOTETomek	Ratio = auto	Random_state = 42	–

5.2 Parameter Tuned for Sampling and Classification Methods

The parameters, namely sampling strategy, randomization, and size of the neighborhood, are fine-tuned, corresponding to over, under, and hybrid sampling, respectively, and discussed in Table 6. The parameter sampling strategy is set to 'not majority' for the oversampling of minority samples. To the contrary, the sampling strategy is set to 'not minority' for the under-sampling of the majority instances. A hybrid sampling procedure both over and under-sample the majority and minority samples simultaneously, the sampling strategy is thus set to 'auto.' The value of the random state is set to 42 to obtain the similar result each time. The value of the nearest neighbor is set to 5, which is used to create samples.

To evaluate the impact of machine learning methods in predicting the publisher's behavior, experiments are performed using 12 machine learning classifiers with tuned parameters for effective working. The details of parameters tuned during experimentation are shown in Table 7.

Table 7 Tuned parameters for classification methods

Methods	Parameter setting
Gradient tree boosting (GTB)	Learning_rate = 0.6, min_samples_split = 100, min_samples_leaf = 5, max_depth = 8, max_features = 'sqrt', subsample = 0.8, random_state = 42,
Random forest (RF)	criterion = entropy, max_depth = 3, n_estimators = 12, random_state = 42
Naïve bayes (NB)	Priors = None, var_smoothing = 1e-09
Logistic regression (LR)	penalty = l1, $C = 1$, random_state = 42
k-nearest neighbor (K-NN)	n_neighbors = 3, p = 2, weights = 'distance'
Decision tree (DT)	criterion = "gini", random_state = 42, max_depth = 5, min_samples_leaf = 4
Support vector machine (SVM)	C = 1, kernel = 'linear', degree = 3, gamma = 1, random_state = 42
Linear discriminant analysis (LDA)	n_components = 5, solver = 'svd', store_covariance = False, tol = 0.0001
AdaBoost + DT (ADT)	n_estimators = 100, base_estimator = dt, learning_rate = 0.1
AdaBoost + SVM (ASM)	n_estimators = 100, base_estimator = svc, learning_rate = 0.1
AdaBoost + LR (ALR)	n_estimators = 100, base_estimator = lr, learning_rate = 0.1
AdaBoost + NB (ANB)	n_estimators = 100, base_estimator = gnb, learning_rate = 0.1

5.3 Classification Results

Experimentations conducted in the current work are an extension of our previous work [12] where twelve standard machine learning classifiers predict the publisher's conduct. For the fair evaluation of the proposed architecture, experiments are conducted on the same datasets and classification methods employed in our previous work. We have overcome the limitation of our previous work, where class imbalance hinders the classification performance. Skewed class distribution is balanced by designing a stacked generalization architecture that balances the class distribution effectively while removing noisy instances. In addition, tenfold cross-validation is used for experimentation on real-world benchmark datasets.

Experiments for stacked generalization are conducted in two steps. In the first step, results are compiled employing architecture-1, which is designed for the resampling of the dataset. For data resampling, data compiled by five oversampling methods are used as input by the hybrid sampling method, resulting in a balanced dataset. In the second step, the balanced dataset is then fed as training data to the eleven popular machine learning base learners of architecture-2, using whose predictions the meta-learner produces the final prediction. The proposed approach is evaluated by employing ten-fold cross-validation on every dataset. The tenfold cross-validation approach divides the imbalanced data into a set of 10. After that, the designed model is trained on 9 sets, and testing is performed using the rest 1 set. The process is repeated ten times, and average predictive results are achieved for the model. The results obtained by the proposed stacked generalization architecture are shown in Tables 8, 9, 10, where C1 - C11 represents the result obtained by base classifiers, while results achieved by meta classifier is indicated in bold. The comparison of

Table 8 Comparing the performance of base and meta classifiers on different datasets based on average precision

Data	Generalizers											
	Base classifiers											Meta
	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
D1	60	59.4	53.5	53.1	57.5	38	41.8	53.3	55.2	56.1	54.5	**66.3**
D2	46.1	33.7	34.4	34	87.3	38.8	34.2	50.2	34.2	54.4	86.9	**85.8**
D3	68.4	75.9	60.1	76.7	78.1	72.2	55.1	60	73.7	77	78.1	**91.7**
D4	53.2	51.5	52.2	51.8	52.6	54.8	52	63.2	54	55.2	52.1	**65.7**
D5	72.5	64.8	82.3	93.7	78.7	82.5	87.8	85	82.9	77.1	73.8	**97.2**
D6	41.8	47.1	71.5	68.8	68.2	61.8	72.2	71.1	40.3	72.7	76.8	**72.6**
D7	69.6	81.7	63.5	77.3	83.1	70	66.5	78.3	71.9	53.7	77.4	**90.2**
D8	63.5	57.8	62.5	61.3	50.2	63	84.7	61.4	63.4	83.8	78.5	**86**
D9	74.1	75.6	74	88.3	75.5	76.8	77.2	73.8	47.1	50.8	81.2	**91**

Note*- C1 - C11 represents the result obtained by base classifiers, while results achieved by meta classifier is indicated in bold

C1: RF, C2: NB, C3: LR, C4: k-NN, C5: DT, C6: SVM, C7: LDA, C8: ADT, C9: ASM, C10: ALR, C11: ANB, C12: GTB

performances of base and meta classifiers based on average precision, recall, and F1-score on different datasets is visually represented in Fig. 5, 6, 7.

GTB, as a Meta-learner, has outperformed all other base classifiers with a precision score of 66.3%. Generation of multiple trees is performed by both Random Forest and Decision Trees towards developing a model. Results demonstrate the

Table 9 Comparing the performance of base and meta classifiers on different datasets based on recall

Data	Generalizers											
	Base classifiers											Meta
	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
D1	52.3	55.4	49.1	47.8	54.4	33.5	40.1	46.9	48.7	49.4	48.8	**62.6**
D2	42.9	36.5	37.2	36.8	87.3	57.1	37	53.5	37	60.2	82.8	**89.7**
D3	69.9	67.2	55.2	78	78.1	96	52.5	57.4	69.2	76.1	71.3	**86.6**
D4	55	53.2	53.9	53.3	52.6	60.9	53.7	65.7	55.7	55	53.3	**59.2**
D5	64.3	50.9	78.4	94.2	78.7	83.6	87.8	85.3	85.4	74.7	71.8	**96.7**
D6	40.1	52.6	63.3	66.8	68.2	54	61.6	65.1	38.4	70.1	73.5	**69.9**
D7	71.2	74.7	57.5	78.8	83.1	74.8	68.2	78.3	74.5	49.3	77.4	**88**
D8	66.5	61	66	59.4	50.2	66.7	88	63.9	66.9	80.5	87	**87.7**
D9	70.5	71.7	70.7	87.6	75.5	79.5	75.7	70.9	43.9	52.9	79.5	**91.5**

Note*- C1 - C11 represents the result obtained by base classifiers, while results achieved by meta classifier is indicated in bold

C1: RF, C2: NB, C3: LR, C4: k-NN, C5: DT, C6: SVM, C7: LDA, C8: ADT, C9: ASM, C10: ALR, C11: ANB, C12: GTB

Table 10 Comparing the performance of base and meta classifiers on different datasets based on F1-score

Data	Generalizers											
	Base classifiers											Meta
	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
D1	55.9	57.3	51.2	50.3	55.9	35.6	40.9	50.3	51.7	52.5	51.5	**63.9**
D2	44.2	35	35.7	35.3	94	22.2	35.5	44.1	30.4	56.7	83.6	**87.5**
D3	69.1	70.2	56.7	77.3	78.1	73.2	53.9	58.5	70.6	76.5	72.6	**89.5**
D4	54	52.3	53	52.5	53.2	20.7	52.8	60.3	52.8	67.4	52.7	**60.9**
D5	65.6	44.8	77.4	93.9	77.1	81.9	87.8	83.5	79.1	76.5	67.2	**97**
D6	40.9	45.4	63.6	67.5	68.4	28.9	61.3	66.1	39.5	47.4	74.7	**70.6**
D7	66.8	76.4	54.7	77.9	83.1	64.5	67	78.3	70.8	51.2	77.4	**89**
D8	64.9	59.1	64.2	60.4	51.3	64.7	86.3	62.8	65.2	86.6	79.4	**86.6**
D9	71.6	73	71.7	87.9	71.9	77.7	76.3	72	45.2	49.4	80.3	**91.3**

Note*- C1 - C11 represents the result obtained by base classifiers, while results achieved by meta classifier is indicated in bold

C1: RF, C2: NB, C3: LR, C4: k-NN, C5: DT, C6: SVM, C7: LDA, C8: ADT, C9: ASM, C10: ALR, C11: ANB, C12: GTB

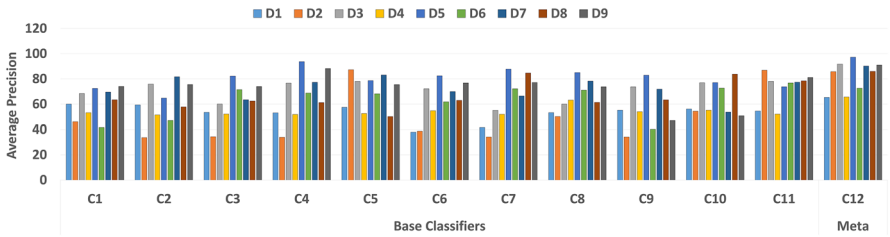

Fig. 5 Performance comparison of base and meta classifiers based on average precision

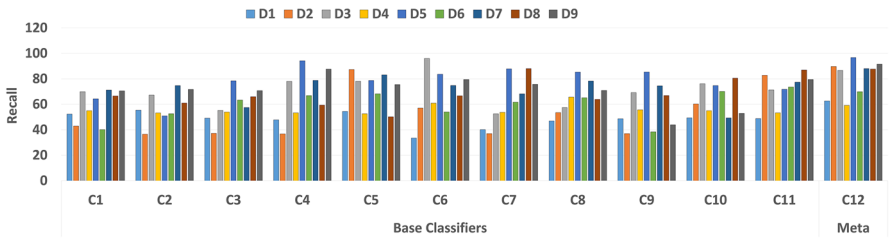

Fig. 6 Performance comparison of base and meta classifiers based onRecall

superiority of RF over DT in predicting the conduct of publishers due to robustness. Random Forest combines multiple decision trees to avert over-fitting, and the error caused due to the bias and enhanced the classification than Decision Tree. Contrary

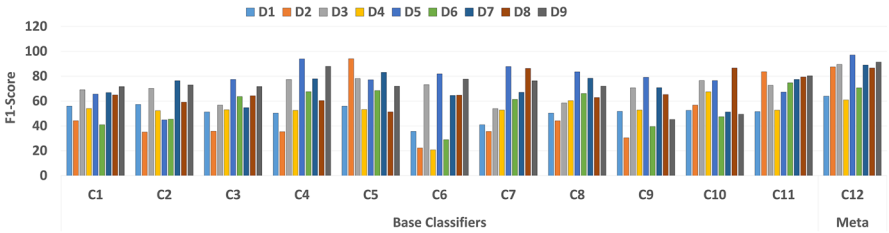

Fig. 7 Performance comparison of base and meta classifiers based on F1-Score

to Adaboost, Gradient Tree Boosting provides minimum weights to instances while attempting to lessen the loss function by getting the mean of the outputs generated by the weak learning models. Gradient Tree Boosting surpasses *k*-NN and Logistic Regression. *k*-NN performed poorly during dealing with high-dimensional datasets and requires distance function towards computing the similarity score, while Logistic Regression underperforms with non-linear decision boundaries. Gradient Tree Boosting has overcome the issues by alleviating the bias by combining several weak classifiers. It develops cascading of predictors, utilizing the output as an input or another classifier. The major strength of Gradient Tree Boosting is that it generates an individual tree simultaneously, with each newly constructed tree aiding in the error correction of the earlier trained tree. Tables 8, 9, 10 shows that Gradient Tree Boosting demonstrates enhanced classification results in precision, recall, and f-measure with appropriate parameter settings on the FDMA2012 user click dataset and eight other unbalanced datasets using the proposed stacked generalization architecture.

6 Discussion

Conventional classification methods generally get biased towards the majority class. Consequentially, the minority class instances are more often misclassified than the majority class instances. Another crucial problem with imbalance is some disjuncts in the dataset, which raises the difficulty of several classification methods during the detection. Current work proposes a framework based on stacked generalization, a procedure that minimizes the generalized error rate of generalizers. Stacked generalized architecture functions by reducing the generalizer's bias concerning a learning dataset. This reduction continues via generalization in another space where inputs are the presumption of generalizers when trained with a segment of a learning dataset and attempts to guess the remaining, which results in an accurate guess. When employed with several generalizers (models), stacked generalization is observed as cross-validation's refined form, utilizing an approach better than crude winner-takes-all of the cross-validation to combine a single generalizer. When employed with an individual generalizer, a stacked generalization mechanism estimates and corrects the error of generalizers trained on a specific learning set. By developing a segment of a learning set, training on a segment of a partition, and observing behaviour on

Table 11 Performance comparison of proposed architecture with previous literature based on resampling

Author	Sampling method with classifier	Result (%)
Berrar [68]	Random forest	36.20
Berrar [17]	Up/down sampling using RF	49.99
Perera et al. [7]	SMOTE	51.40
Vasumati et al.[69]	Random forest	52.30
Oentaryo et al.[25]	Random forest	58.84
Sisodia et al. [12]	GTB	60.50
Sisodia et al.[21]	Bagging + DT	63.92
Taneja et.al [16]	SMOTE + HDDT	64.00
Sisodia et.al [15]	ADASYN + GTB	64.32
Our contribution	Stacked generalization with GTB	66.3

other segments, the designed architectures have reduced the bias of generalizers concerning a learning set. The designed stacked generalization resampling architecture effectively balances the class distribution while making the data ambiguous-free by removing those samples closer to each other and making the classification difficult. For further evaluation of the effectiveness of the designed work, the performance of the proposed architecture is compared with previous literature on FDMA 2012 dataset, which is demonstrated in Table 11.

7 Conclusion

In this work, we designed a framework comprising stacked generalization architecture for resampling and classification that aids in the identification of fraudulent publishers. The base models of the first architecture were trained on an imbalanced dataset whose predictions were then fed as an input to the meta-model, which resulted in a balanced dataset as the final prediction. Furthermore, the base models of the second architecture were then trained over the final predicted balanced dataset obtained from the first architecture. The predicted results obtained from level-0 generalizers of the second architecture were then provided to the meta-learner for final prediction. Experimentation was conducted to evaluate stacked generalized architecture on the user-click dataset. Experimentation was performed to obtain a balanced dataset.

In contrast, the final set of predictions was obtained over the balanced dataset, demonstrating how the stacked generalization improved the classification performance in predicting fraudulent publishers. The results have empirically prove that the proposed architecture has accurately rebalanced the class distribution of imbalanced datasets. It has also improved the generalizability of learning models using the potentials of state-of-the-art generalizers at each level to allow each generalizer to benefit from the other generalizer's expertise. The publisher's prediction and classification improvement are achieved with meta learner using GTB with an average precision score of 66.3% on FDMA 2012 user-click data. The work concludes that the stacked generalization resampling (SGR) architecture was an effective method

that improvises the learning behaviour of classifiers when dealing with skewed datasets, as it helped the learning approaches from the risk of getting biased.

Limitation and Future Work: The limitation of the work lies in the more memory required by the stacked models. Moreover, the training time in these stacked models is higher than in other state-of-the-art models. In future, these issues need to be addressed while deploying the stacked models in production.

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Declarations

Conflict of interest All authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any authors.

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